



Medical Policy

Subject: Axial Lumbar Interbody Fusion

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Description/Scope

This document addresses axial or presacral lumbar interbody fusion, a minimally invasive technique in which anterior access to the L4-S1 disc spaces is used for interbody fusion to minimize damage to muscular, ligamentous, neural, and vascular structures. It is performed under fluoroscopic guidance using specialized instrumentation.

Note: Please see the following related documents for additional information:

- [SURG.00071 Percutaneous Spinal Surgery](#)

Note: For a high-level overview of this document, please see "[Summary for Members and Families](#)" below

Position Statement

Investigational and Not Medically Necessary:

Axial or presacral lumbar interbody fusion is considered **investigational and not medically necessary**.

Summary for Members and Families

This document describes clinical studies and expert recommendations, and explains why we don't consider axial lumbar interbody fusion to be clinically appropriate. The following summary does not replace the medical necessity criteria or other information in this document. The summary may not contain all of the relevant criteria or information. This summary is not medical advice. Please check with your healthcare provider for any advice about your health.

Key Information

Axial lumbar interbody fusion (AxiaLIF®) is a type of spine surgery used to treat lower back problems. It is done through a small incision near the tailbone and uses real-time x-ray guidance to reach the correct part of the spine. A device and bone graft are placed to help two bones in the lower spine grow together, which is called fusion. This method aims to reduce damage to nearby muscles and tissues compared to traditional surgery. However, AxiaLIF has not been proven to improve health compared to other standard treatments. There are also risks, including injury to the bowel, blood vessels, or nerves, infection, and exposure to radiation during the procedure.

What the Studies Show

Most studies on this procedure are small or do not compare this treatment directly with other options in a strong way. Some studies report that people had less pain and better function after surgery. However, many studies did not include people treated with more standard treatments, so it is unclear whether improvements seen after lumbar interbody fusion were at least as good as improvements seen after standard treatments. Better studies are needed to know if this treatment improves health compared to other spine surgeries.

Some studies have compared this type of spine fusion to other fusion methods. Some found it works about the same, while others found it may be less successful in helping the bones fully join together. Reviews that looked at many studies found that this procedure does not appear to work better than standard methods. Medical experts also report that there is no clear proof that this approach leads to better results. This procedure also has risks. These can include injury to the bowel, infection, exposure to x-ray radiation, and other complications. Because the available evidence is limited and not strong, it is not clear if the benefits outweigh the risks.

Is this clinically appropriate?

This procedure is not appropriate because it has not been proven to improve health. Studies are mostly small, do not include comparison groups, and may be biased. Some results suggest improvement, but without strong comparisons, it is unclear if this method is better than other treatments. Better studies are needed to know if axial lumbar interbody fusion improves health. Using treatments that are not proven exposes people to risks without health benefits. Axial lumbar interbody fusion is not clinically appropriate.

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Rationale

Summary

Axial lumbar interbody fusion, also known as presacral or trans-sacral fusion, is a minimally invasive procedure that uses a percutaneous paracoccygeal approach under fluoroscopic guidance to access the L4-S1 or L5-S1 disc space for interbody fusion. The evidence base is limited to retrospective case series, technical reports, and systematic reviews, with no randomized controlled trials (RCTs) or robust comparative studies. Although some studies report improvements in pain, function, and fusion rates, these findings are derived from uncontrolled studies and may be subject to bias. Comparative evidence suggests that outcomes may be no better than standard techniques. Guideline statements including a statement from the American Association of Neurological Surgeons (AANS, 2014) state no conclusive evidence favoring specific interbody fusion approaches. The overall evidence is insufficient to demonstrate improved health outcomes. Due to the variable natural history of the disorder and subjective nature of outcomes such as pain, the available uncontrolled studies do not allow conclusions about whether axial LIF is as beneficial as other surgical approaches to lumbosacral interbody fusion.

Discussion

The published studies evaluating axial lumbar interbody fusion (axial LIF) are limited to technical reports (Gerszten, 2012; Lindley, 2011; Marchi, 2012), case series (Balsano, 2020; Melgar, 2014; Michael, 2019; Patil, 2010; Tobler, 2013; Tobler, 2011; Whang, 2013; Zeilstra, 2013), or systematic reviews of the available studies (Schroeder, 2015; Schroeder, 2016). Case series lack control or comparison groups, and many of them were retrospective and/or had relatively short-term follow-up. No published RCTs or other prospective controlled studies evaluating the efficacy and safety of axial LIF were identified.

A 2025 case series (Korytkowski, 2025) evaluated AxiaLIF as a “salvage” option for six individuals with complex spine conditions for whom standard fusion approaches were not feasible due to prior surgeries, comorbidities, or high surgical risk. The study reported generally favorable technical outcomes, including stable implants and the absence of major intraoperative complications, with some postoperative issues such as infection and wound complications resolving with treatment. The evidence is limited by its very small sample size, single-center retrospective design, lack of a control group, and highly selected population. This evidence is categorized as Level IV with limited generalizability and an inability to form meaningful comparisons to standard fusion techniques. The findings are insufficient to establish safety and effectiveness.

Whang (2013) retrospectively compared the radiographic fusion rates and adverse events in 96 individuals who underwent L5-S1 interbody fusion using either a standard anterior retroperitoneal approach or the AxiaLIF system with supplemental posterior fixation. Fusion success at 24 months was assessed using multiplanar computed tomography scans reviewed by two independent observers with a 4-point grading scale. Arthrodesis rates were 79% for anterior lumbar interbody fusion (ALIF) and 85% for AxiaLIF ($p>0.05$). The number and types of adverse events were similar between groups, with one serious intraoperative complication, iliac artery laceration, reported in the ALIF group. However, differences in adjunctive graft materials may have influenced results, as more individuals in the AxiaLIF group received recombinant growth factors compared to the ALIF group (29 vs. 11), which may account for the higher fusion rate observed with AxiaLIF.

Michael (2019) retrospectively reviewed medical records of 149 individuals who had undergone two-level axial LIF and had at least 2 years of follow-up. The mean duration of follow-up was 6 years. A total of 20 individuals (13.4%) developed adjacent-segment disease (ASD) during follow-up. Kaplan-Meier estimates of disease-free ASD survival rate were 95.3% (95% confidence interval [CI], 90.4% to 97.7%) at 2 years and 89.1% (95% CI, 82.8% to 93.2%) at 5 years after two-level fusion.

In 2020, Balsano published retrospective data on 52 individuals who were treated with AxiaLIF. Diagnoses included L5 isthmic spondylolisthesis, low-grade dysplasia, primary degenerative disc disease, and disc disease secondary to previous discectomy. Data on pain assessed by a visual analogue scale (VAS) were reported for 43 individuals who had 2 years of follow-up. The mean VAS score at baseline was 7.8. This decreased significantly to 2.3 at 12 months and 1.7 at 24 months. Similarly, scores on the Oswestry Disability Index (ODI), which was 0.502 at baseline, improved significantly to a mean of 0.168 at 12 months and 0.138 at 24 months. As is true for other case series, it is not possible to draw conclusions about causality due to the lack of a comparison group.

Results of an industry-sponsored post-marketing study reporting on complications with AxiaLIF were reported by Gundanna in 2011. The study reported on a database of 9152 individuals who underwent interbody fusion with the AxiaLIF device. A total of 120 complications (1.3%) were reported, 54% of which occurred within 5 days of surgery. The most commonly reported complication was bowel injury (n=59, 0.6%), followed by transient intraoperative hypotension (n=20, 0.2%). All other complications had an incidence of 0.1% or lower.

Schroeder published two systematic reviews. The 2015 systematic review identified 15 publications on axial interbody arthrodesis of the L5-S1 spine using the AxiaLIF device, 13 case series and 2 retrospective cohort studies. Based primarily on the case series, the authors reported a high overall fusion rate (93.15%) and a complication rate of 12.9% associated with axial interbody arthrodesis. However, due to the limited prospective data, the actual fusion rates may be lower and complication rates may be higher than reported in the studies.

In 2016, Schroeder published a systematic review comparing fusion rates after ALIF, transforaminal lumbar interbody fusion (TLIF), and axial LIF at the lumbosacral junction in adults undergoing surgery for one- and two-level degenerative spine conditions. A total of 42 articles and 1507 study participants were included in the review. A difference in overall fusion rates was identified, with a rate of 99.2% (range, 96.4%-99.8%) for TLIF, 97.2% (range, 91.0%-99.2%) for ALIF, and 90.5% (range, 79.0%-97.0%) for axial interbody fusion (p=0.005). In a paired analysis directly comparing fusion techniques, only the difference between a TLIF and an axial interbody fusion was statistically significant. No statistically significant difference between the three techniques was identified when bilateral pedicle screws supported the interbody fusion (p>0.05). A limitation of this review includes a paucity of RCTs directly comparing the techniques. Additionally, of the reviewed studies, only 12 were found to have a low risk of bias. There was significant heterogeneity in how a solid fusion was determined (that is, use of computed tomography versus radiographs). Confounding variables not accounted for in this review included adequacy of the endplate preparation and medical comorbidities.

In 2014, the AANS (Mummaneni, 2014) published guidelines on fusion procedures for degenerative disease of the lumbar spine, stating, "There is no conclusive evidence demonstrating improved clinical or radiographic outcome based on the different interbody fusion techniques."

Background/Overview

Axial LIF is a percutaneous technique utilizing a paracoccygeal approach and trans-sacral instrumentation to stabilize the L4 to S1 or L5 to S1 spinal segment(s) that has been proposed as a method of achieving fusion with reduced complications when compared to open spinal fusion surgery. The AxiaLIF® and subsequent variations such as the AxiaLIF® II or 2-Level Systems (TranS1®, Houston, TX) were cleared for marketing through the U.S. Food and Drug Administration (FDA) 510(k) process. In the original 510(k) clearance document, AxiaLIF was determined to be substantially equivalent to a previously cleared spinal fixation system and no clinical data on the AxiaLIF were reported (FDA, 2004).

The AxiaLIF and AxiaLIF II Level Systems consist of techniques and surgical instruments for creating a pre-sacral access route to perform percutaneous fusion of the L5-S1 or L4-S1 vertebral bodies. The procedure utilizes fluoroscopic guidance for a blunt guide introducer that is passed through a 15-20 millimeter (mm) incision lateral to the coccyx and advanced along the midline of the anterior surface of the sacrum. A guide pin is introduced and tapped into the sacrum. A series of graduated dilators are passed along the guide pin to open a working channel for the passage of instruments. After debulking the nucleus pulposus, bone graft material is injected to fill the disc space. A threaded rod designed to restore disc and neural foramen height is then secured in place. This procedure can be performed at two levels.

FDA documents state that the procedures are intended to provide anterior stabilization of the spinal segments as an adjunct to spinal fusion and for assisting in the treatment of degeneration of the lumbar disc, performing lumbar discectomy, or for assistance in the performance of interbody fusion. The AxiaLIF Systems are indicated for use in individuals requiring fusion to treat pseudoarthrosis, unsuccessful previous fusion, spinal stenosis, spondylolisthesis (Grade 1), or degenerative disc disease as defined as back pain of discogenic origin with degeneration of the disc confirmed by history and radiographic studies. The technique is not intended to treat severe scoliosis, severe spondylolisthesis (Grades 2, 3 and 4), tumor, or trauma. In addition, the AxiaLIF is not intended for use in individuals with vertebral compression fractures or other conditions where the mechanical integrity of the vertebral body is compromised. Use of axial lumbar interbody fusion is limited to anterior supplemental fixation of the lumbar spine at L4-S1 or L5-S1 in conjunction with legally marketed facet or pedicle screw systems.

Complications after an axial LIF procedure may include perforation of the bowel and injury to blood vessels and/or nerves as well as infection (Shen, 2007). Since the procedure uses fluoroscopic guidance, the length of a procedure can expose the individual to high doses of radiation.

Definitions

Anterior: The front surface of the body.

Axial skeleton (as related to the human body): Is comprised of the vertebral column, the spine and much of the skull.

Fluoroscopy: Imaging technique to obtain real-time moving images of the internal structures of the body; this imaging uses an x-ray source and fluorescent screen; modern fluoroscopes couple the screen to an x-ray image intensifier and video camera allowing the images to be recorded and shown on a monitor.

Presacral: Anterior to the sacrum.

Spondylolisthesis: A forward dislocation of one vertebra over the one beneath it producing pressure on spinal nerves

Coding

The following codes for treatments and procedures applicable to this document are included below for informational purposes. Inclusion or exclusion of a procedure, diagnosis or device code(s) does not constitute or imply member coverage or provider reimbursement policy. Please refer to the member's contract benefits in effect at the time of service to determine coverage or non-coverage of these services as it applies to an individual member.

When services are Investigational and Not Medically Necessary:

CPT

22586	Arthrodesis, pre-sacral interbody technique, including disc space preparation, discectomy, with posterior instrumentation, with image guidance, includes bone graft when performed, L5-S1 interspace
22899	Unlisted procedure, spine [when specified as pre-sacral interbody arthrodesis lumbar. L4-L5 interspace with instrumentation, or pre-sacral interbody arthrodesis L4-L5 or L5-S1 interspace without instrumentation]

ICD-10 Procedure

0SG03A0	Fusion of lumbar vertebral joint with interbody fusion device, anterior approach, anterior column, percutaneous approach
0SG13A0	Fusion of 2 or more lumbar vertebral joints with interbody fusion device, anterior approach, anterior column, percutaneous approach
0SG33A0	Fusion of lumbosacral joint with interbody fusion device, anterior approach, anterior column, percutaneous approach

ICD-10 Diagnosis

All diagnoses

References

Peer Reviewed Publications:

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6. Marchi L, Oliveira L, Coutinho E, Pimenta L. Results and complications after 2-level axial lumbar interbody fusion with a minimum 2-year follow-up. *J Neurosurg Spine.* 2012; 17(3):187-192.
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12. Schroeder GD, Kepler CK, Vaccaro AR. Axial interbody arthrodesis of the L5-S1 segment: a systematic review of the literature. *J Neurosurg Spine*. 2015; 23(3):314-319.
13. Shen FH, Samartzis D, Khanna AJ, et al. Minimally invasive techniques for lumbar interbody fusion. *Orthop Clin N Am*. 2007; 38(3):373-386.
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16. Whang PG, Sasso RC, Patel VV, et al. Comparison of axial and anterior interbody fusions of the L5-S1 segment: a retrospective cohort analysis. *J Spinal Disord Tech*. 2013; 26(8):437-443.
17. Zeilstra DJ, Miller LE, Block JE. Axial lumbar interbody fusion: a 6-year single-center experience. *Clin Interv Aging*. 2013; 8:1063-1069.

Government Agency, Medical Society, and Other Authoritative Publications:

1. Mummaneni PV, Dhall SS, Eck JC, et al. Guideline update for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 11: Interbody techniques for lumbar fusion. *J Neurosurg Spine*. 2014; 21(1):67-74.
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Axial LIF
 AxiaLIF I
 AxiaLIF II
 AxiaLIF+
 AxiaLIF System

The use of specific product names is illustrative only. It is not intended to be a recommendation of one product over another, and is not intended to represent a complete listing of all products available.

Document History

Status	Date	Action
Reviewed	05/14/2026	Medical Policy and Technology Assessment Committee (MPTAC) review. Added "Summary of Members and Families section." Revised Description, Rationale, Background and References sections.
Reviewed	05/08/2025	MPTAC review. Revised References section.
Reviewed	05/09/2024	MPTAC review. Rationale and References sections updated.
Reviewed	05/11/2023	MPTAC review. References section updated.
Reviewed	05/12/2022	MPTAC review. References section updated.
Reviewed	05/13/2021	MPTAC review. Rationale, Background/Overview and References sections updated.
Reviewed	05/14/2020	MPTAC review. Rationale, Background/Overview and References sections updated.
Reviewed	11/07/2019	MPTAC review. Rationale and References sections updated.
Reviewed	01/24/2019	MPTAC review. Rationale, References and Index sections updated.
	12/27/2018	Updated Coding section with 01/01/2019 CPT changes; removed 0195T and 0196T deleted 12/31/2018.
Reviewed	03/22/2018	MPTAC review. Updated Description, Rationale, Background, References and Index sections.
	12/27/2017	The document header wording updated from "Current Effective Date" to "Publish Date." Updated Coding section with 01/01/2018 CPT changes; removed 0309T deleted 12/31/2017, added 22899.
Reviewed	05/04/2017	MPTAC review. Updated Rationale and References sections.
Reviewed	05/05/2016	MPTAC review. Updated Rationale and References sections. Removed ICD-9 codes from Coding section.
Reviewed	05/07/2015	MPTAC review. Updated Rationale and References sections. Format changes throughout document.
Reviewed	05/15/2014	MPTAC review. Updated Rationale, Background, and References sections.
Reviewed	05/09/2013	MPTAC review. Updated Rationale, References, and Index.
	01/01/2013	Updated Coding section with 01/01/2013 CPT changes.
Reviewed	05/10/2012	MPTAC review. Rationale and References updated.
Reviewed	05/19/2011	MPTAC review. Rationale and References updated.

Reviewed	05/13/2010	MPTAC review. References updated.
New	05/21/2009	MPTAC review. Initial document development.

Federal and State law, as well as contract language, including definitions and specific contract provisions/exclusions, take precedence over Medical Policy and must be considered first in determining eligibility for coverage. The member's contract benefits in effect on the date that services are rendered must be used. Medical Policy, which addresses medical efficacy, should be considered before utilizing medical opinion in adjudication. Medical technology is constantly evolving, and we reserve the right to review and update Medical Policy periodically.

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